

Please check the examination details below before entering your candidate information

Candidate surname					Other names				
Centre Number					Candidate Number				

Pearson Edexcel International Advanced Level

Thursday 16 January 2025

Morning (Time: 1 hour 30 minutes) **Paper reference** **WMA13/01**

Mathematics

International Advanced Level

Pure Mathematics P3

You must have:
Mathematical Formulae and Statistical Tables (Yellow), calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 10 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- If you change your mind about an answer, cross it out and put your new answer and any working underneath.

Turn over ►

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1. $f(x) = 2\sec x + 6x - 3$ $0 < x < \frac{\pi}{2}$

The equation $f(x) = 0$ has a single root α

(a) Show that $0.1 < \alpha < 0.2$

(2)

(b) Show that α is a solution of

$$x = \frac{1}{2} - \frac{1}{3\cos x}$$

(1)

The iterative formula

$$x_{n+1} = \frac{1}{2} - \frac{1}{3\cos x_n}$$

is used to find α

(c) Starting with $x_1 = 0.15$ and using the iterative formula,

(i) find, to 4 decimal places, the value of x_2

(ii) find, to 4 decimal places, the value of α

(3)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 6 marks)



2. The weed on the surface of a pond is being monitored.

The surface area of the pond covered by the weed, $A \text{ m}^2$, is modelled by the equation

$$\log_{10} A = 1 + 0.03t$$

where t is the number of weeks after monitoring began.

Use the equation of the model to answer parts (a) and (b).

- (a) Find the surface area of the pond initially covered by the weed.

(1)

After T weeks, 25 m^2 of the pond is covered by the weed.

- (b) Find the value of T , giving your answer to 2 decimal places.

(2)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 3 marks)



3.

In this question you must show all stages of your working.**Solutions relying on calculator technology are not acceptable.**

A curve has equation

$$y = \frac{4x + 1}{(x + 3)^2} \quad x \neq -3 \quad x \in \mathbb{R}$$

Use calculus to find the range of values of x for which y is increasing.

(6)



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Question 3 continued

Handwriting practice area with horizontal lines.

(Total for Question 3 is 6 marks)



4. Given that

$$\frac{4x^3 + 2x^2 + 3x + 8}{x^2 + 4} \equiv Ax + B + \frac{Cx + D}{x^2 + 4}$$

(a) (i) find the values of the constants A , B and C

(ii) show that $D = 0$

(4)

(b) Hence, using algebraic integration, find

$$\int_1^4 \frac{4x^3 + 2x^2 + 3x + 8}{x^2 + 4} dx$$

giving your answer in the form $p + q \ln 2$, where p and q are integers.

(5)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 9 marks)



5. A hot piece of metal is cooled by dropping it into water. The temperature, $H^{\circ}\text{C}$, of the metal, t minutes after it is dropped into the water, is modelled by the equation

$$H = 280e^{-0.05t} + 24 \quad t \geq 0$$

Use the equation of the model to answer parts (a) to (d).

- (a) Find the initial temperature of the piece of metal. (1)
- (b) On Diagram 1, sketch the graph of H against t . On your sketch, state the equation of the asymptote to the curve. (2)
- (c) Find the value of t for which $H = 144$, giving your answer to 2 decimal places.
- (Solutions based entirely on calculator technology are not acceptable.)* (3)
- (d) Show by differentiation that

$$\frac{dH}{dt} = a + bH$$

where a and b are constants to be found.



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Question 5 continued

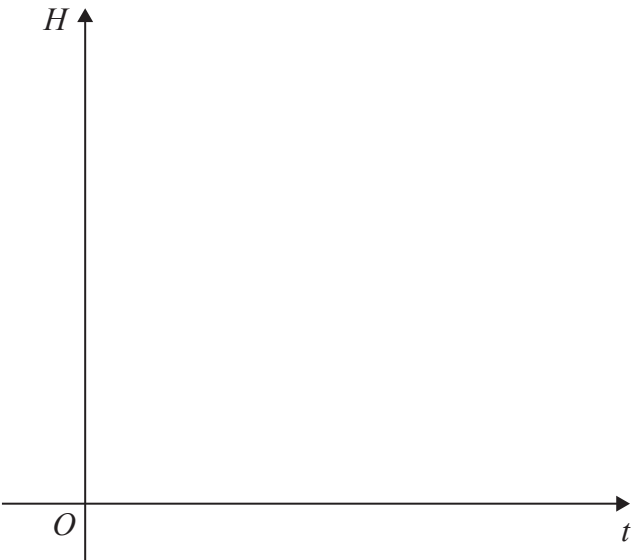


Diagram 1

Lined area for writing the answer.

Question 5 continued

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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 9 marks)



- 6.** The function f is defined by

$$f(x) = \frac{4x + 3}{x - 2} \quad x \neq 2$$

- (a) Find f^{-1} (3)

- (b) Show that

$$\text{ff}(x) = \frac{ax + b}{cx + d}$$

where a, b, c and d are integers to be found. (3)

The point $P(3, 15)$ lies on the curve with equation $y = f(x)$.

- (c) Find the point to which P is mapped when $y = f(x)$ is transformed to the curve with equation $y = 2f(3x) + 8$
- (2)**



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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

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(Total for Question 6 is 8 marks)



7. Given that a and b are positive constants with $a > b$,

(a) sketch, on **separate** diagrams, the graph with equation

(i) $y = |3x - a|$

(ii) $y = |3x - a| - b$

Show on each sketch

- the coordinates of the minimum point on the graph
- the coordinates of the point at which the graph crosses the y -axis

(6)

(b) Solve the equation

$$|3x - a| - b = 5x$$

giving any solution for x in terms of a and b .

(2)



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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 8 marks)



8.

In this question you must show all stages of your working.**Solutions relying entirely on calculator technology are not acceptable.**

- (i) Solve, for
- $0 < \theta < \pi$

$$3 \operatorname{cosec} \theta = 8 \cos \theta$$

giving your answers, in radians, to 3 significant figures.

(5)

- (ii) Solve, for
- $0 < x < 180^\circ$

$$\frac{\tan 2x - \tan 70^\circ}{1 + \tan 2x \tan 70^\circ} = -\frac{3}{8}$$

giving your answers, in degrees, to one decimal place.

(4)



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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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Question 8 continued

Handwriting practice area with horizontal lines.

(Total for Question 8 is 9 marks)



9.

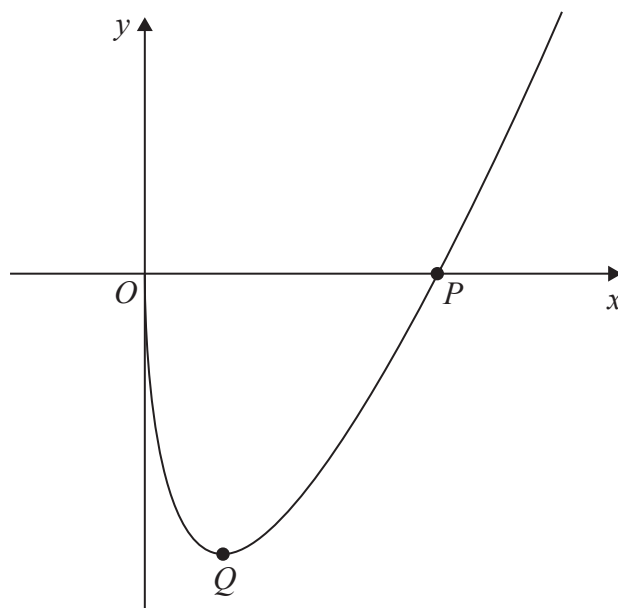


Figure 1

Figure 1 shows a sketch of part of the curve C with equation $y = f(x)$ where

$$f(x) = 6\sqrt{x} \ln(4x) \quad x > 0$$

The curve cuts the x -axis at point P

- (a) State the x coordinate of P

(1)

The point Q , shown in Figure 1, is the stationary point on C

- (b) Use calculus to find the exact coordinates of Q

(5)

- (c) Hence find the range of the function $g(x)$ where

$$g(x) = -2f(x)$$

(2)



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Question 9 continued

Lined area for writing the answer to Question 9.



Question 9 continued

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Question 9 continued

Handwriting practice area with horizontal lines.

(Total for Question 9 is 8 marks)



10.

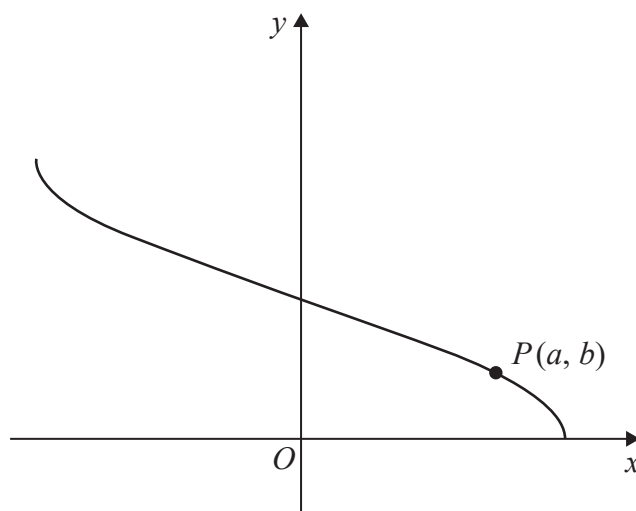


Figure 2

Figure 2 shows a sketch of the curve with equation

$$x = 3 \cos 2y \quad -3 \leq x \leq 3 \quad 0 \leq y \leq \frac{\pi}{2}$$

- (a) Find $\frac{dx}{dy}$ in terms of y .

(2)

- (b) Hence show that

$$\frac{dy}{dx} = \frac{k}{\sqrt{9 - x^2}}$$

where k is a constant to be found.

(3)

The point $P(a, b)$ lies on the curve and is shown in Figure 2.

Given that

- the gradient of the curve at P is $-\frac{1}{4}$
- both a and b are positive

- (c) find the exact values of a and b .

(4)



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Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

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(Total for Question 10 is 9 marks)

TOTAL FOR PAPER IS 75 MARKS

